

# Incorporating health economics into models

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Modular Certificate Course in  
Modelling for Pandemic Preparedness and  
Responses (MPPR)

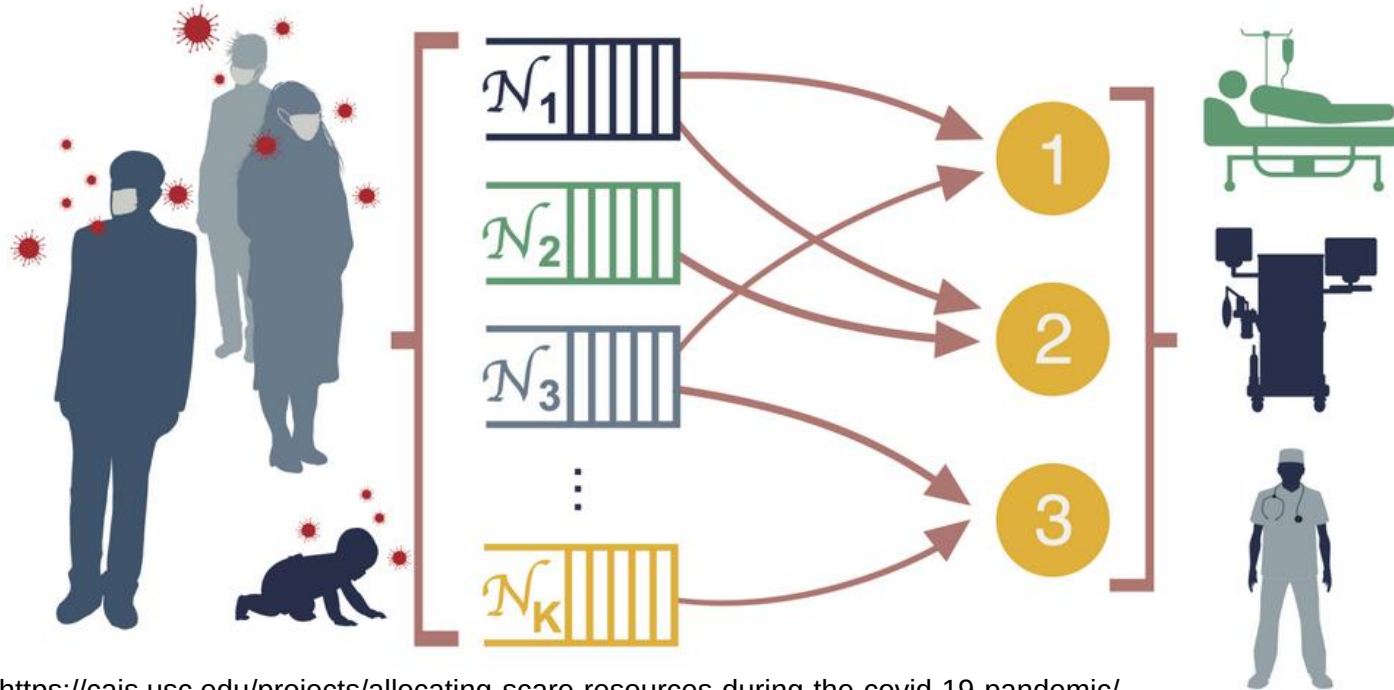
27 Sept 2024



# Questions typically prompted by new pathogen

1. “When should we hit STOP on the economy?”
  2. “How many will die?”
  3. “Where will the next wave hit?”
  4. “How bad will the next wave be?”
  5. “Are we in a new wave yet?”
  6. “How many beds do we need?”
  7. “What restrictions should we implement?”  
“Should we close/ open schools?” “Do we still need masks?” “Can we allow events again?” “Do travel restrictions work?”
  8. “Just how bad is this wholly new variant?”
  9. “What will this cost us?”
  10. “Which drugs/ vaccines should we invest in?”  
Should South Africa join COVAX?”
- Cost-benefit analysis
  - Long-term epi model
  - Spatial analysis
  - Scenario analysis
  - Resurgence metrics
  - Short-term forecasts
  - ? (limited data on single restrictions)
  - Assessing new variants
  - Budget impact analysis
  - Cost-effectiveness analysis

## In economics we are always concerned by scarcity of resources



Source: <https://cais.usc.edu/projects/allocating-scare-resources-during-the-covid-19-pandemic/>

Scarcity means **decisions** need to be made about what interventions should be provided

## Scarcity...

Journal of Critical Care 61 (2021) 76–81



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journal homepage: [www.journals.elsevier.com/journal-of-critical-care](http://www.journals.elsevier.com/journal-of-critical-care)



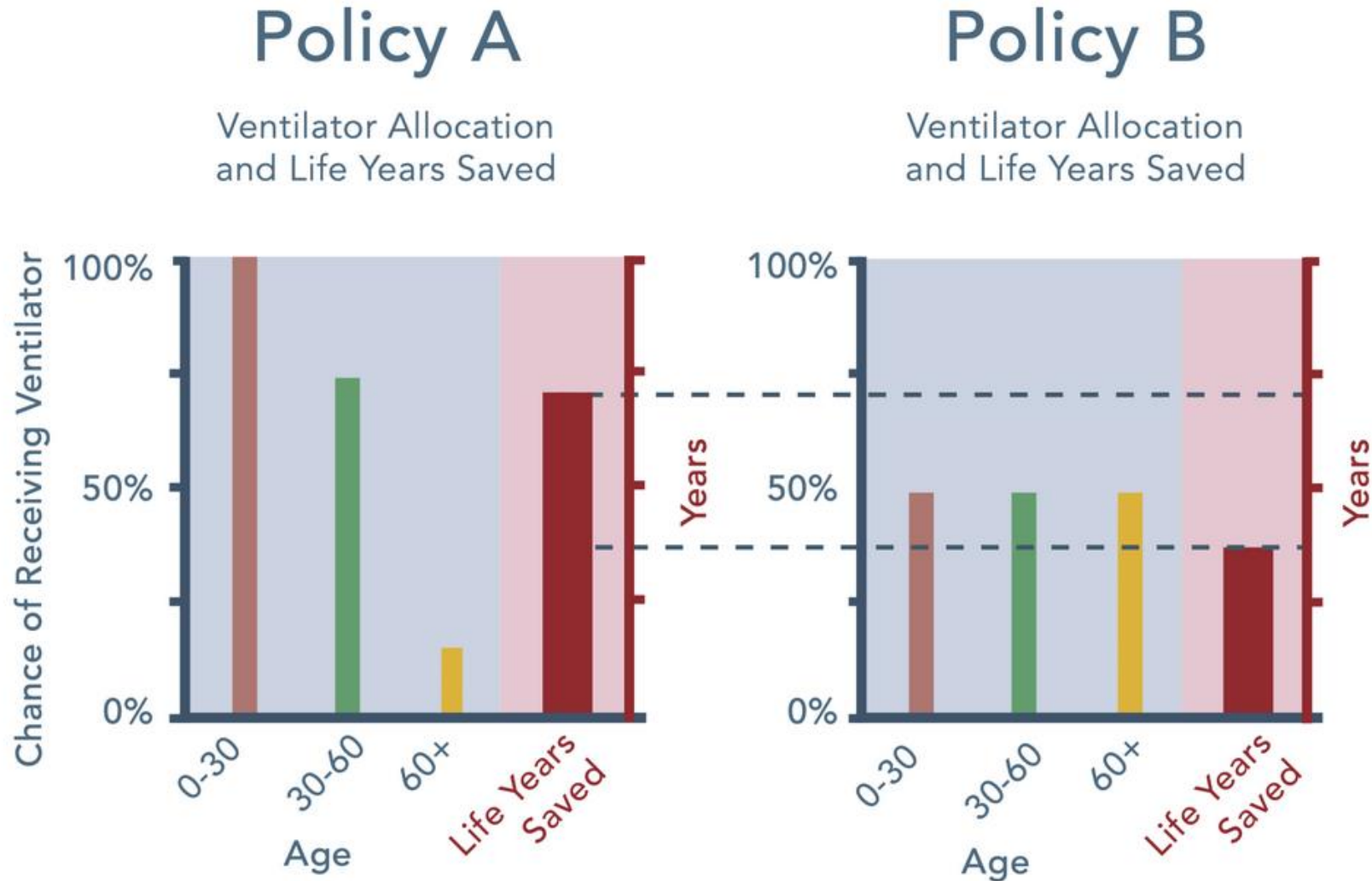
### Capacity of intensive care units in Ghana

Moses Siaw-Frimpong, MD<sup>a</sup>, Sunkaru Touray, MD, MSc, FACP<sup>b</sup>, Nana Sefa, MD, MPH<sup>c,\*</sup>



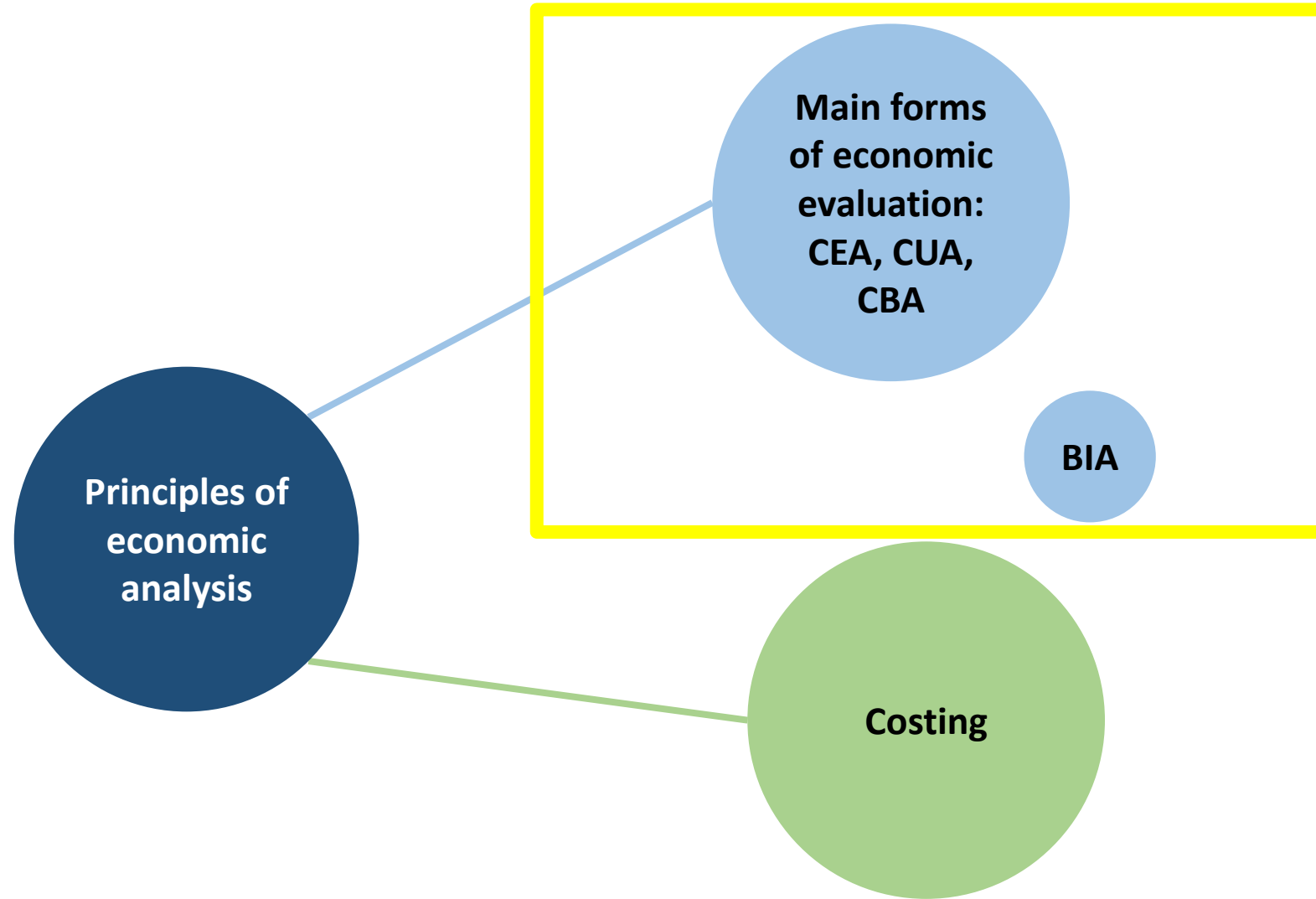
- Cross-sectional observational study of all operating ICUs in Ghana
- 16 operating ICUs in 9 hospitals (13 adult and 3 pediatric ICUs)
- Total of 113 adult and 36 pediatric ICU beds for population of 30 million  
→ 0.5 ICU beds per 100,000 people
- Median 5 staffed ICU beds and 4 ventilators per ward
- **6 adult and 2 pediatric intensivists**
  - 56% of ICUS staffed solely by non-intensivist providers
- Adequate nursing support and essential critical care medications
- Current financing model for critical care delivery creates barrier for most patients

...means decisions need to be made

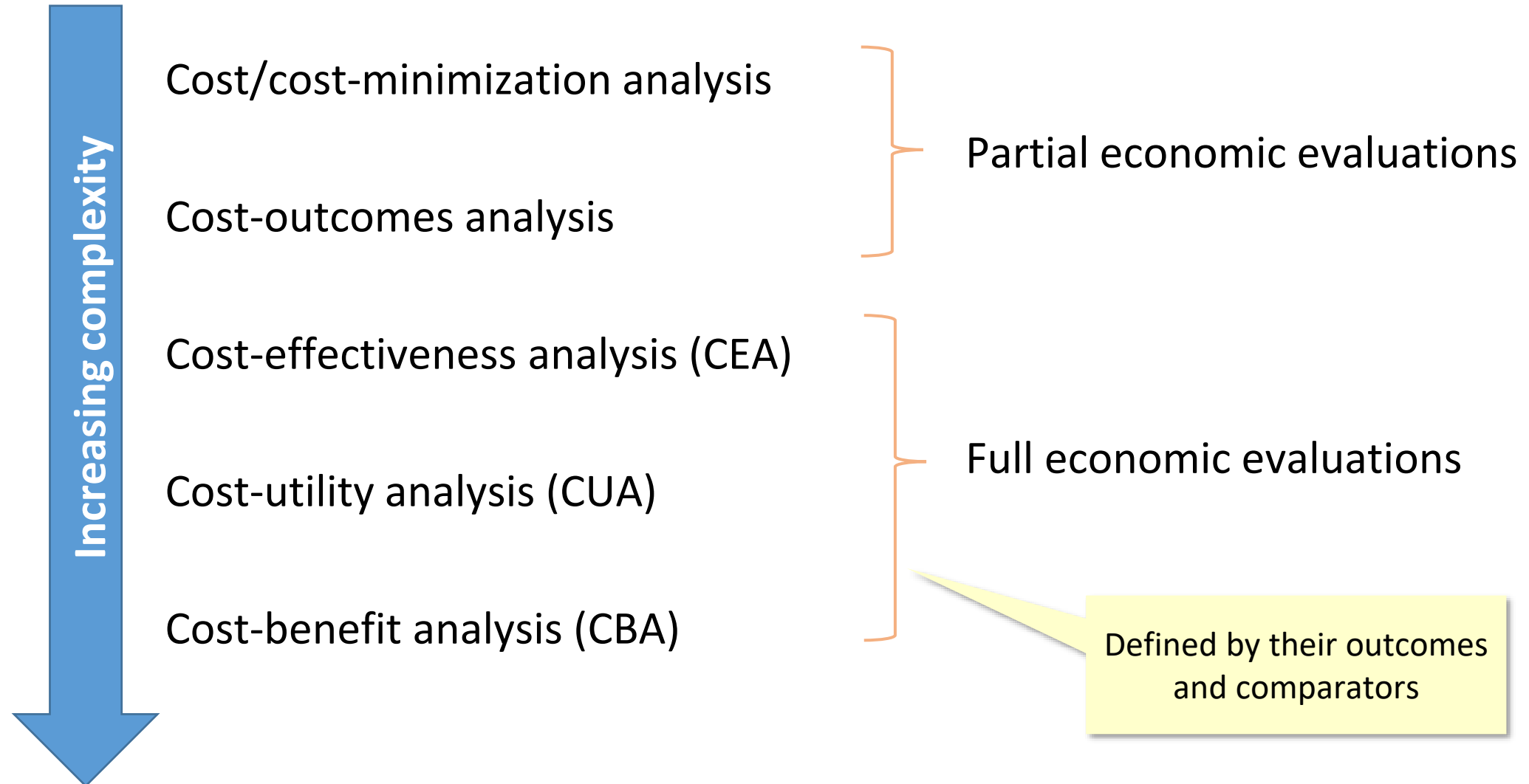


Source: <https://cais.usc.edu/projects/allocating-scare-resources-during-the-covid-19-pandemic/>

# Focus of this session



# Types of economic evaluation



# Measuring outcomes

Increasing complexity

- **Process measure:** measures activities or outputs of the intervention
  - Cost of TB treatment per person year
- **Intermediate outcome:** reflects intermediate changes due to the intervention of the project, before there is a health impact
  - Cost per person *successfully* treated for TB
- **Primary outcome:** measures final effect or impact on health status due to the intervention
  - **Natural outcome:**
    - Cost per TB case averted; cost per life year gained
  - **Compound outcome:**
    - QALY gained, DALY averted

needs to consider secondary and dynamic effects

e.g., impact of TB treatment on TB transmission, of PrEP on condom use (trials, modelling)



**Cost minimization analysis** is relatively simple but can only be used if alternative interventions have **the same outcomes** and the goal is to find the least costly intervention

Alternate interventions considered?



Costs examined?

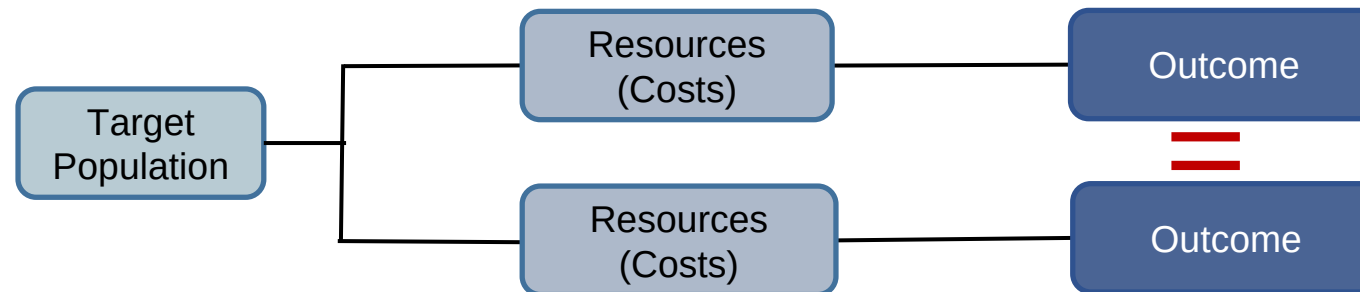


Outcomes examined?



**Example:**

*If vaccine A and vaccine B have the **same clinical effectiveness**, which vaccine will cost less to provide to eligible clients?*



**Key Feature:**

Outcomes are not compared across the interventions, so outcomes must be **exactly the same**

**Cost outcome analysis** is not usually considered as its own evaluation category but it is useful to remember

Alternate interventions considered?



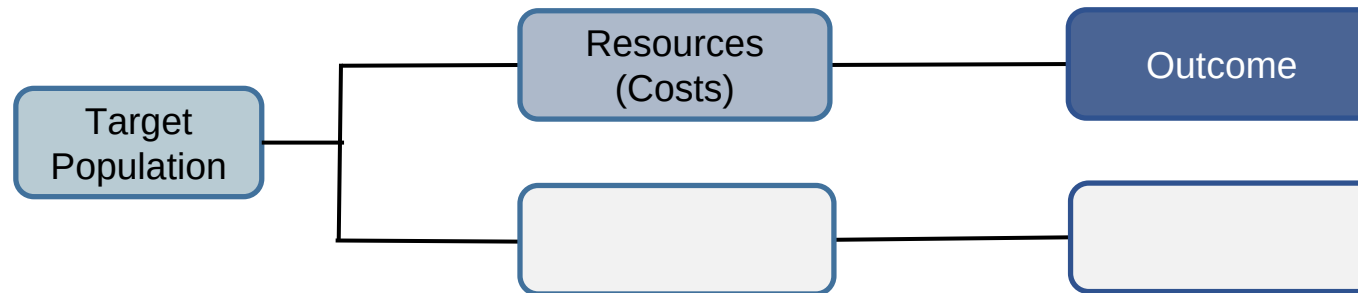
Costs examined?



Outcomes examined?



*What is the cost of Paxlovid  
per COVID-19 hospitalisation  
averted?*



**Key  
Feature:**

There are no alternate interventions being considered so this is not a full economic evaluation

## Cost-effectiveness analysis (CEA) is perhaps the most common economic analysis. It uses **natural or disease specific outcomes**

Alternate interventions considered?



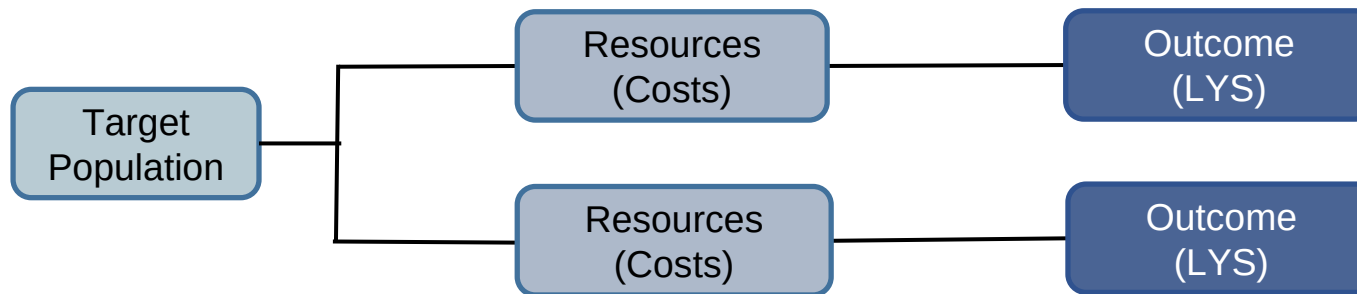
Costs examined?



Outcomes examined?



*If vaccine A and vaccine B have different **clinical effectiveness** and **different prices**, which is more cost effective?*



**Key Feature:**

Multiple interventions are compared on both costs and outcomes. e.g. cost/HIV infection averted

Remember you will have to define your comparator for a CEA, and “do nothing” is a legitimate alternative!

**Cost-utility analysis (CUA)** adds quality of life (=utility) to health outcomes so they are more comparable across disease areas

Alternate interventions considered?



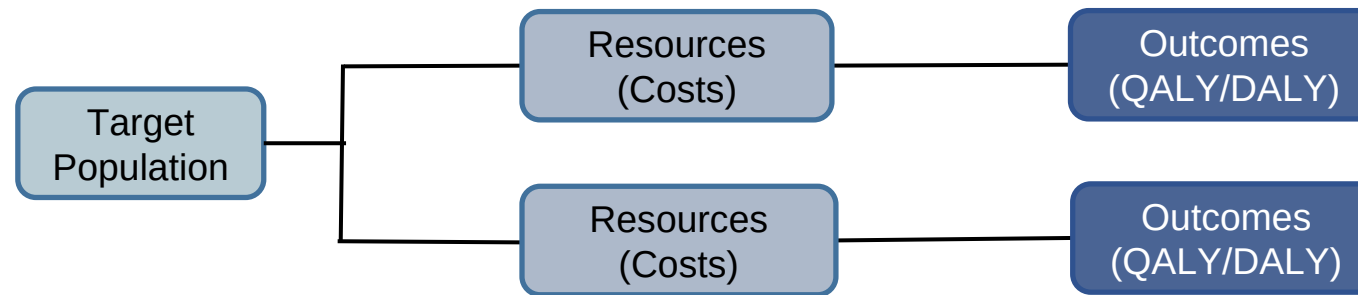
Costs examined?



Outcomes examined?



*Would health gains be maximised by **prioritising COVID-19 booster vaccination or malaria vaccination?***



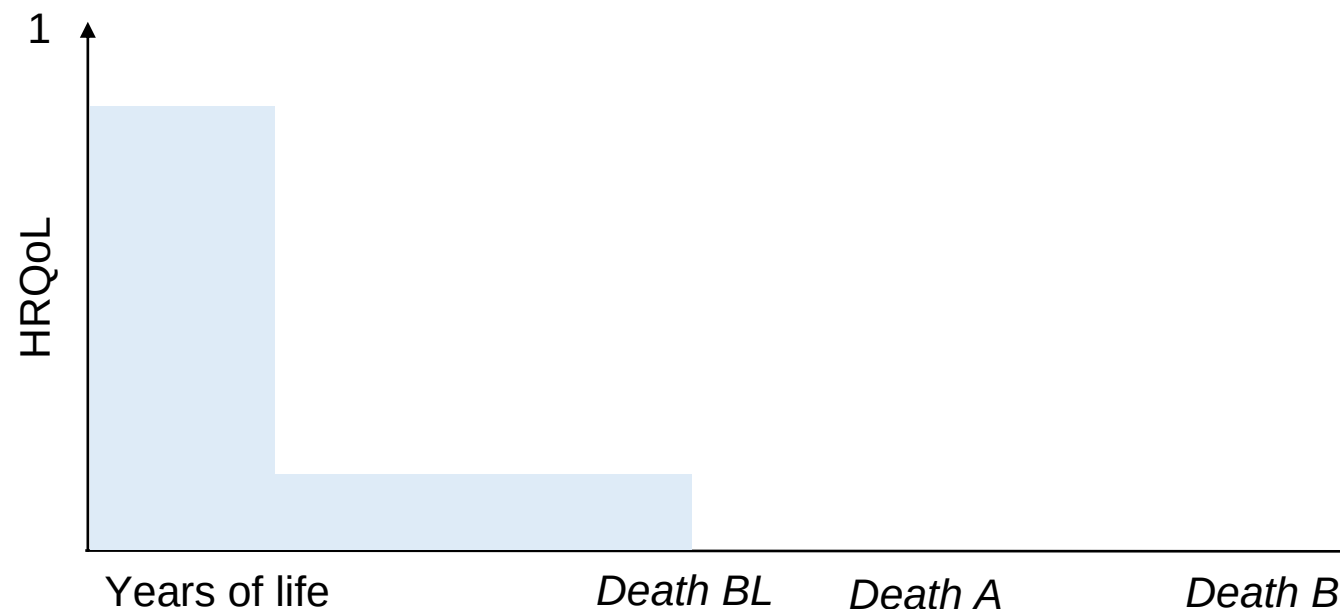
**Key Feature:**

CUA always includes primary (final) outcomes such as *QALYS gained* or *DALYs averted*. This means it can compare interventions across disease areas

## Representing your outputs as QALYs gained makes it more comparable across disease area, but are complicated to produce

### QALY

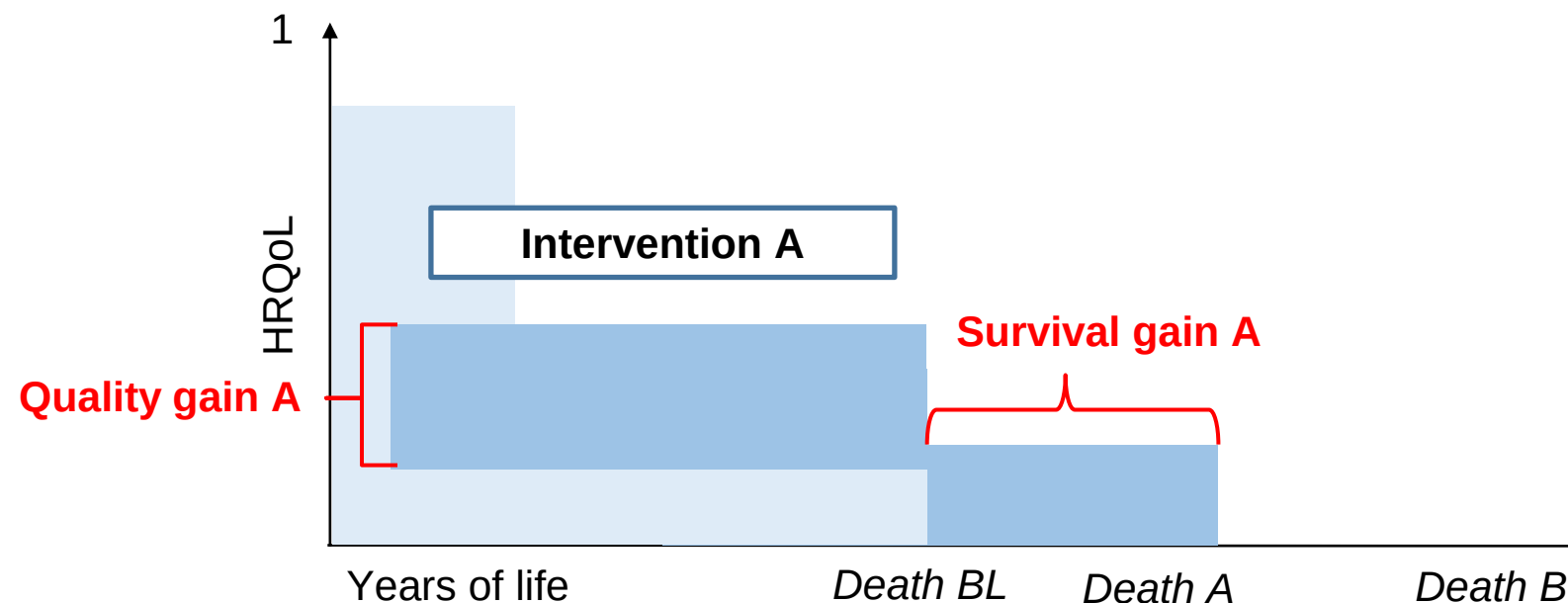
Health is a function of both the length of life, as well as the quality of life.  
A QALY of 1 means a year of life lived in perfect health



## Representing your outputs as QALYs gained makes it more comparable across disease area, but are complicated to produce

### QALY

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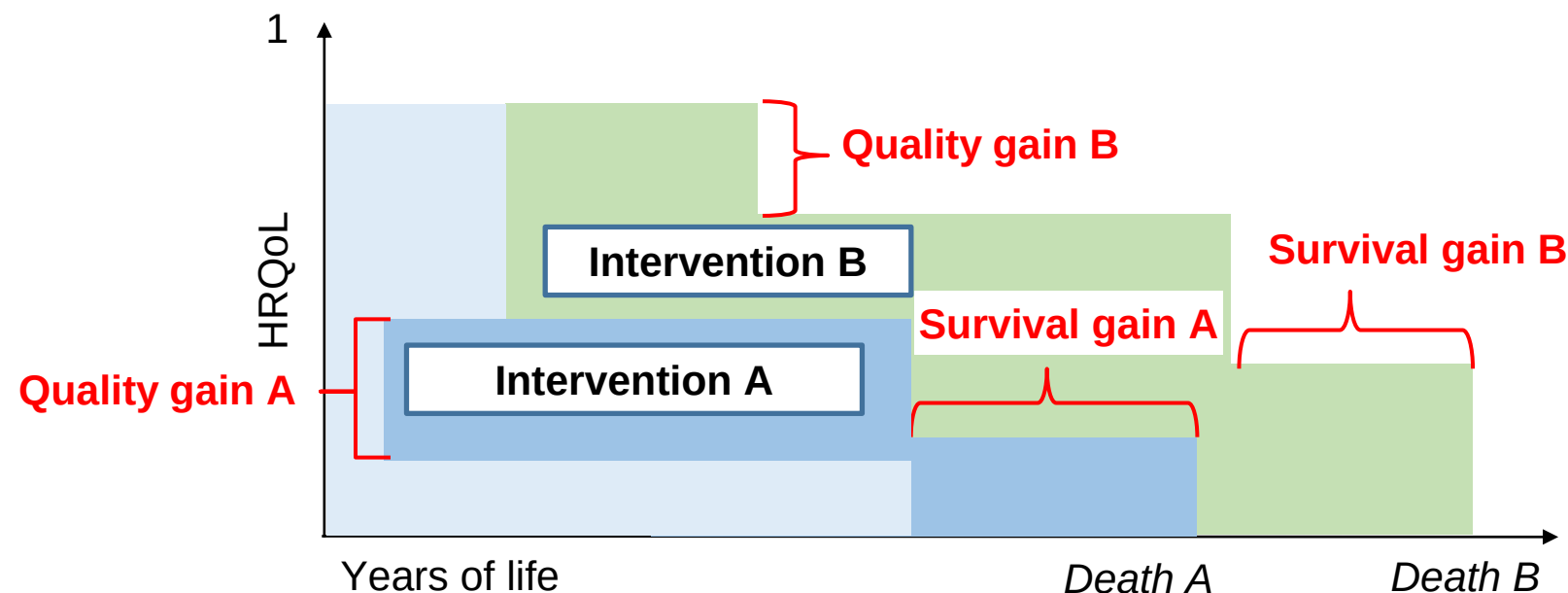


## Representing your outputs as QALYs gained makes it more comparable across disease area, but are complicated to produce

### QALY

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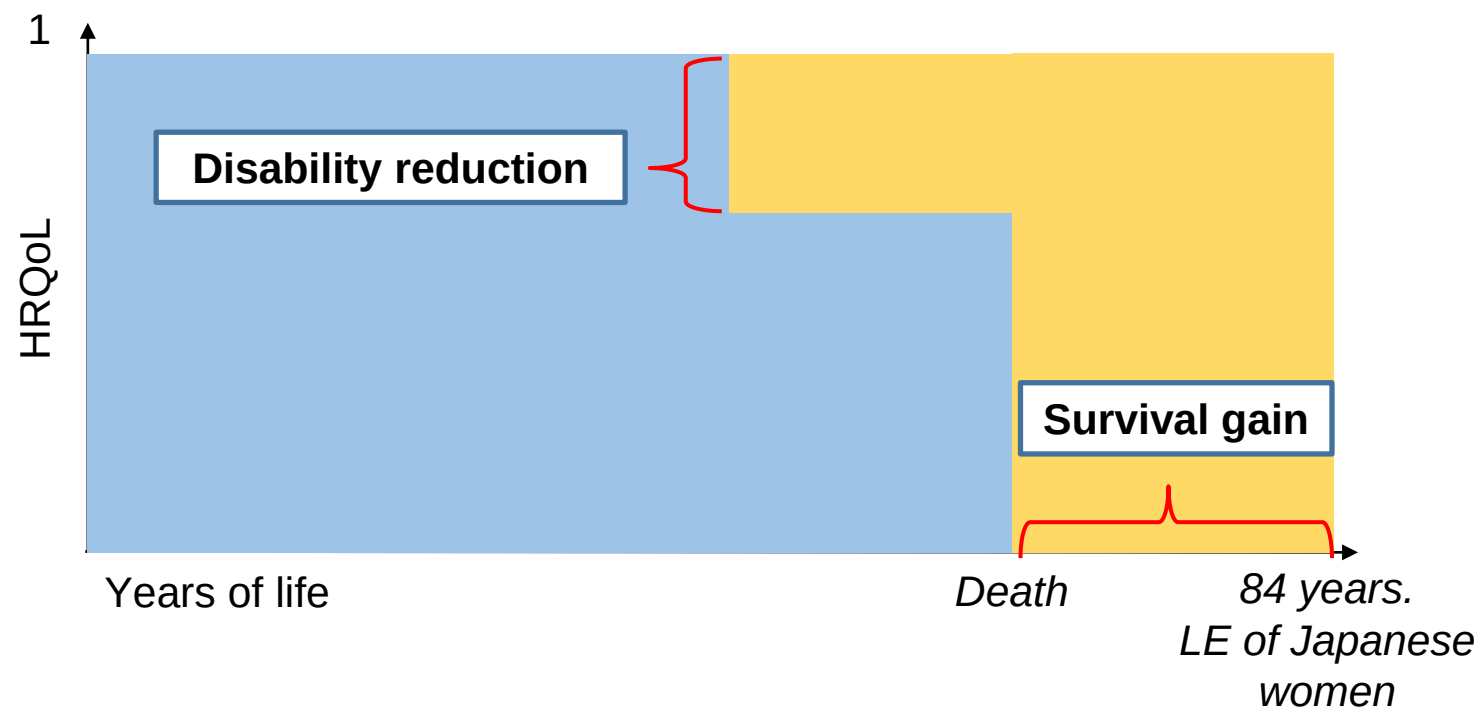


Under intervention B the QALYS gained are equal to the green shaded area, so the goal is to maximise this

**DALYs are based on internationally agreed disability weights, and are designed to measure and compare disease burden across populations**

## DALY

DALYs are calculated from life expectancy, adjusted by early death and the burden of living with a disease or disability



Concerns remain about the utility weights for QALYs and DALYs not being representative of African populations



**CEA produces incremental cost-effectiveness ratios (ICERs) which indicate which of the alternate interventions should be chosen**

$$\text{ICER} = \frac{\text{Cost of new intervention} - \text{cost of old intervention}}{\text{Outcome of new intervention} - \text{outcome of old intervention}}$$

Example:

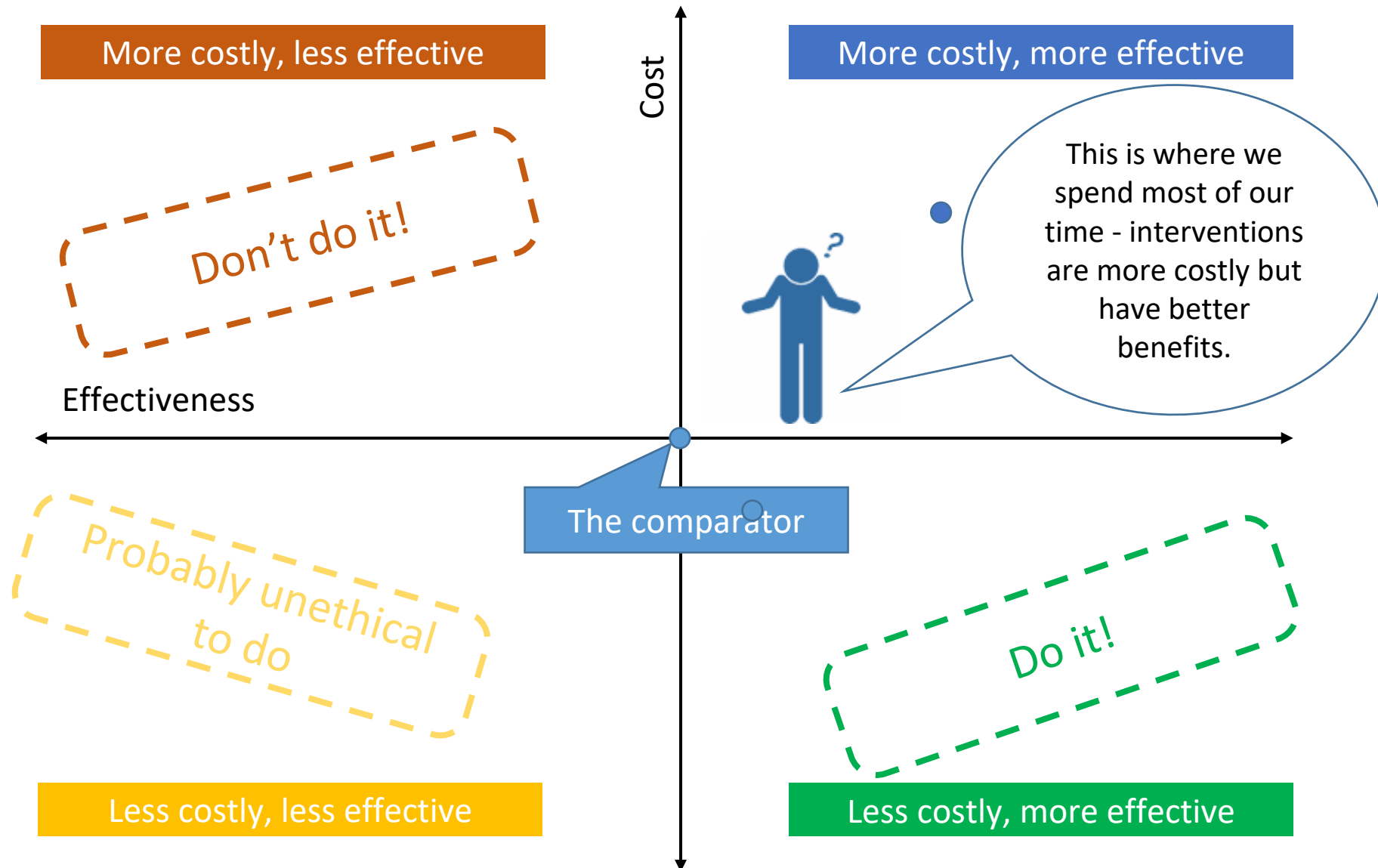
Calculate the ICER:

|                  | <b>Total cost (€)</b> | <b>Outcome (LYS)</b> |
|------------------|-----------------------|----------------------|
| Old intervention | 1 300 000             | 500                  |
| New intervention | 5 000 000             | 900                  |

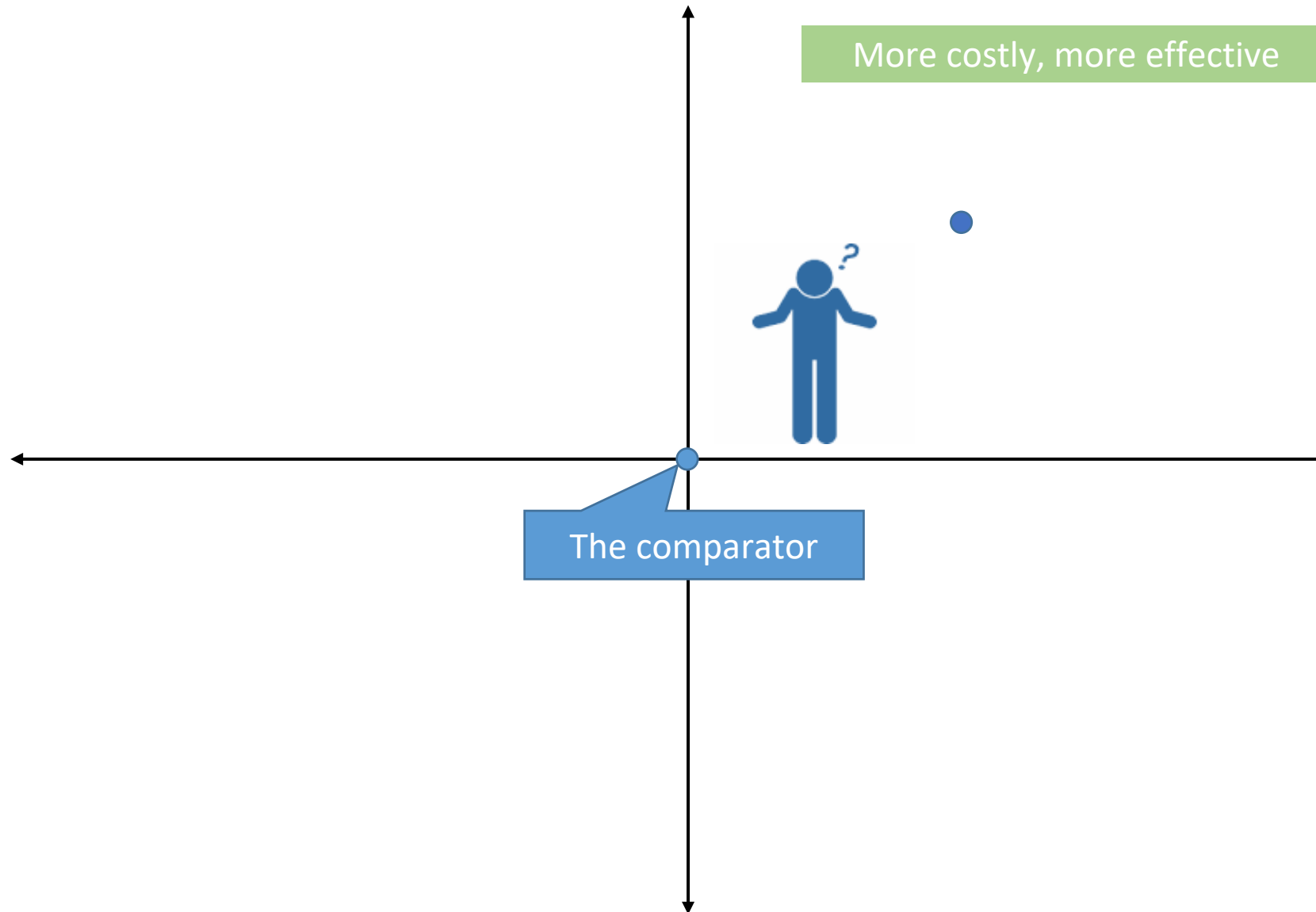
$$\text{ICER} = \text{€ 9 250 per LYS}$$

Next question – **is this cost-effective?**

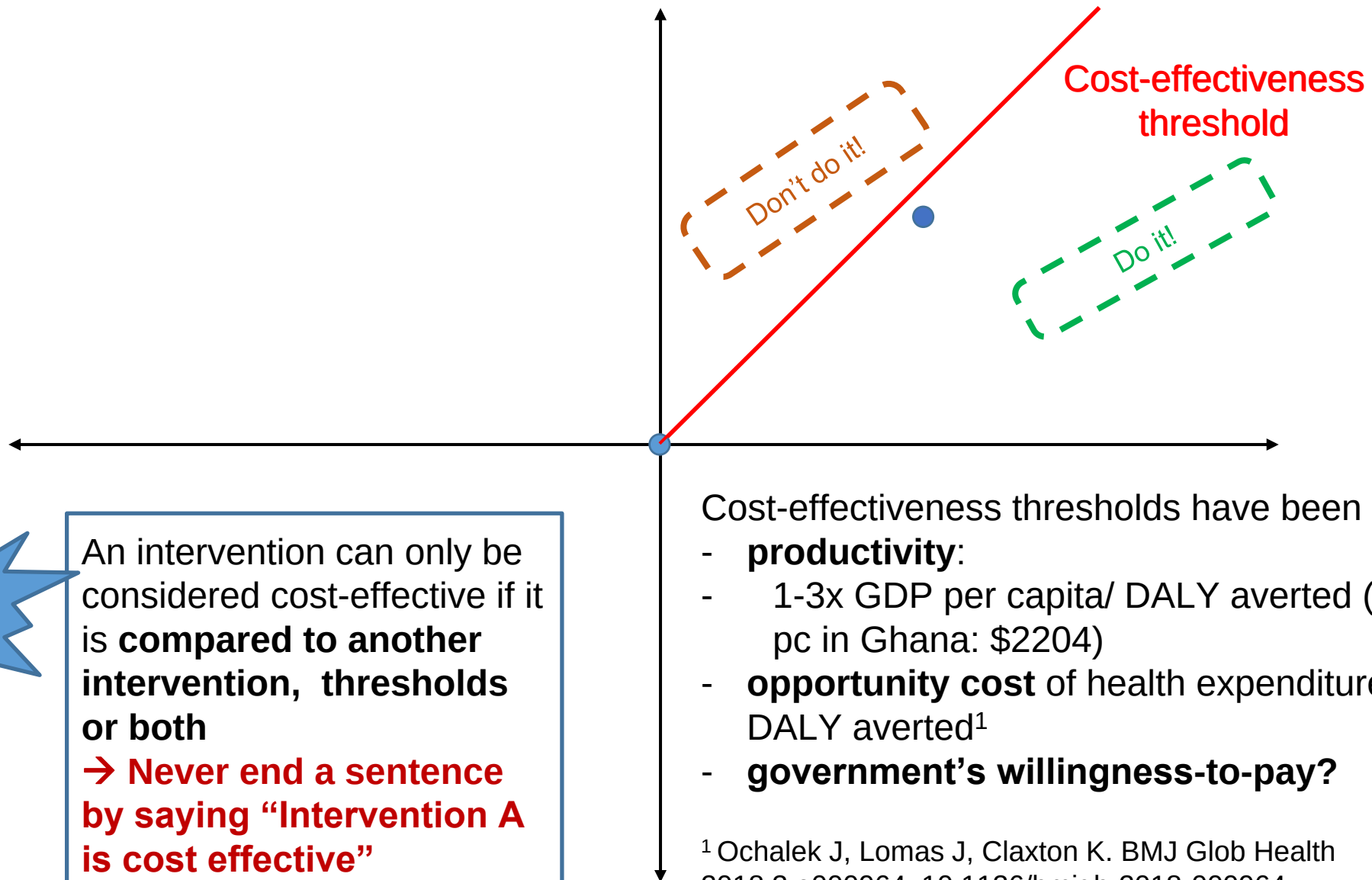
## A cost-effectiveness plane can help make sense of CEAs



## A cost-effectiveness plane can help make sense of CEAs



## A cost-effectiveness plane can help make sense of CEAs





# SACMC examples: Cost effectiveness analyses

## Cost effectiveness analyses of

- remdesivir, tocilizumab, molnupiravir
- all vaccines offered to South Africa (Pfizer, J+J, Sinovac) (*prospective*)
- **vaccine programme (*retrospective*)**
  - incl. Wave 3 (Feb-Nov 2021)
  - incl. Wave 3+4 (Feb 2021-Apr 2022)
  - incremental over baseline of *no policy response*; including hospitalisation costs averted

# CEA of vaccine programme: Results (probabilistic analysis)

**Analysis incl. W3**

|                                                           |                   | Hosp. deaths only | 85% of excess deaths |
|-----------------------------------------------------------|-------------------|-------------------|----------------------|
|                                                           | Total costs       | DALYs lost        |                      |
| No vaccine                                                | \$418,890,680     | 1,183,311         | 2,175,807            |
| Vaccine                                                   | \$956,816,924     | 942,342           | 1,768,919            |
| Difference                                                | Incremental costs | DALYs averted     |                      |
| (Vaccine vs No vaccine)                                   | \$537,926,244     | 240,969           | 406,888              |
| Incremental cost/ DALY averted<br>(Vaccine vs No vaccine) |                   | \$2,232           | \$1,322              |

**CET= USD 2 318/DALY**

# CEA of vaccine programme: Results (probabilistic analysis)

**Analysis incl. W4**

|                                                           |                        | Hosp. deaths only | 85% of excess deaths |
|-----------------------------------------------------------|------------------------|-------------------|----------------------|
|                                                           | Total costs            | DALYs lost        |                      |
| No vaccine                                                | \$607,878,725 (+45%)   | 1,608,114 (+36%)  | 3,246,586 (+49%)     |
| Vaccine                                                   | \$1,349,426,873 (+41%) | 1,209,023 (+28%)  | 2,542,441 (+44%)     |
| Difference                                                | Incremental costs      | DALYs averted     |                      |
| (Vaccine vs No vaccine)                                   | \$741,548,148 (+38%)   | 399,091 (+66%)    | 704,145 (+73%)       |
| Incremental cost/ DALY averted<br>(Vaccine vs No vaccine) |                        | \$1,858 (-17%)    | \$ 1,053 (-20%)      |

**CET= USD 2 318/DALY**

# CEA of vaccine programme: Conclusions

- **Vaccination programme has become more cost effective with inclusion of additional wave**
  - accounted for a wider age group
- Increasing vaccination coverage further will however likely be less cost effective
  - linked to waning of current immunity
- ***Cost effectiveness did not seem to be a major decision criterion in gov't decision making*** (more driven by “rule of rescue” and macro-economic considerations)



## Cost-benefit analysis (CBA) represents outcomes in monetary terms

Alternate interventions considered?



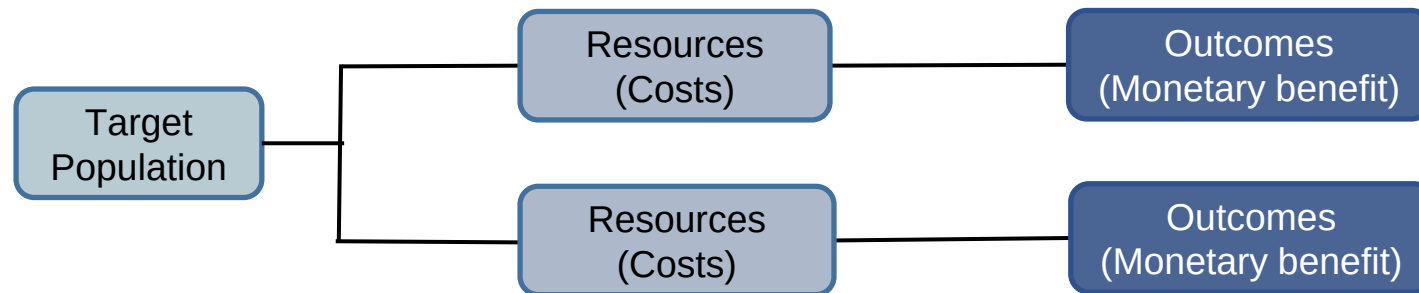
Costs examined?



Outcomes examined?



*Did the **monetary**  
**benefits of the COVID-19**  
**lockdown outweigh the**  
**costs?***



**Key  
Feature:**

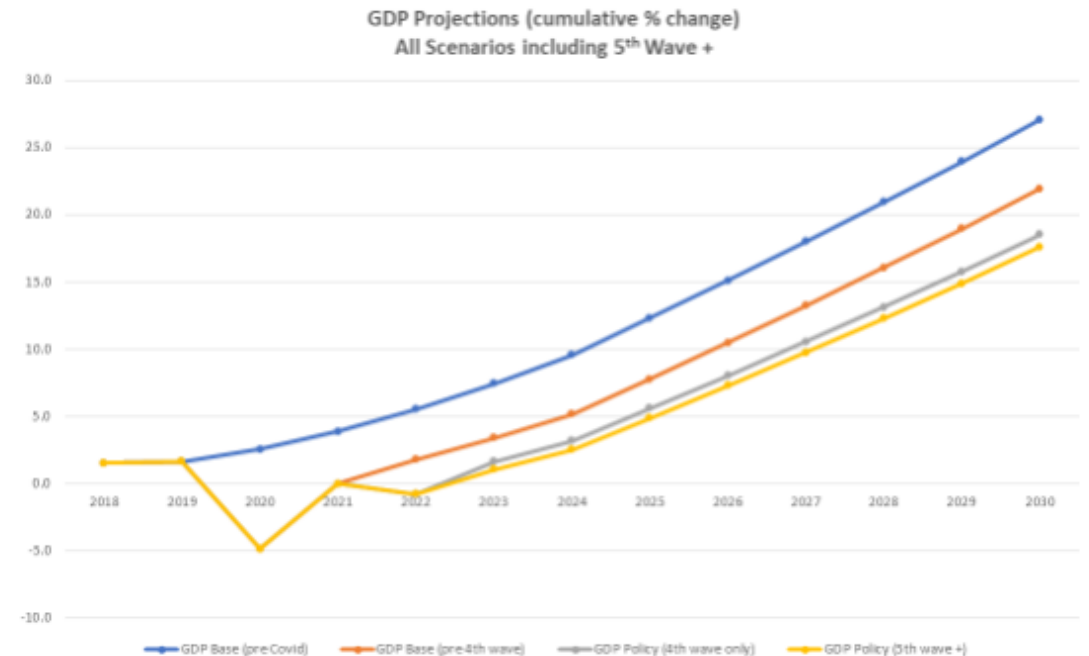
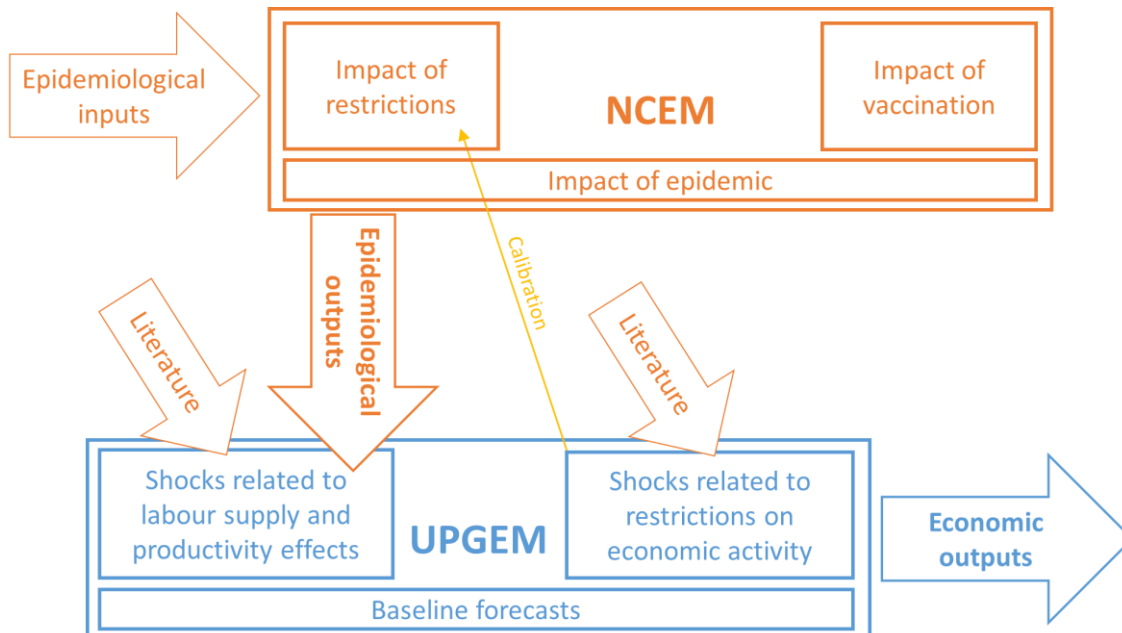
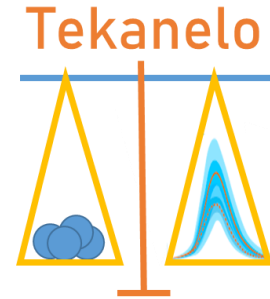
Outcomes presented as monetary values allows for comparison across budget areas, such as health vs education

Estimation of outcomes in monetary terms is complex as has to include productivity, taxation and other monetary benefits from regained health



# SACMC examples: Cost-benefit analysis

## Jointly modelling the epi and macro-economic impact of COVID-19: The Tekanelo Model



## Budget impact analysis (BIA) is important for considering the **affordability** of interventions

Alternate interventions considered?



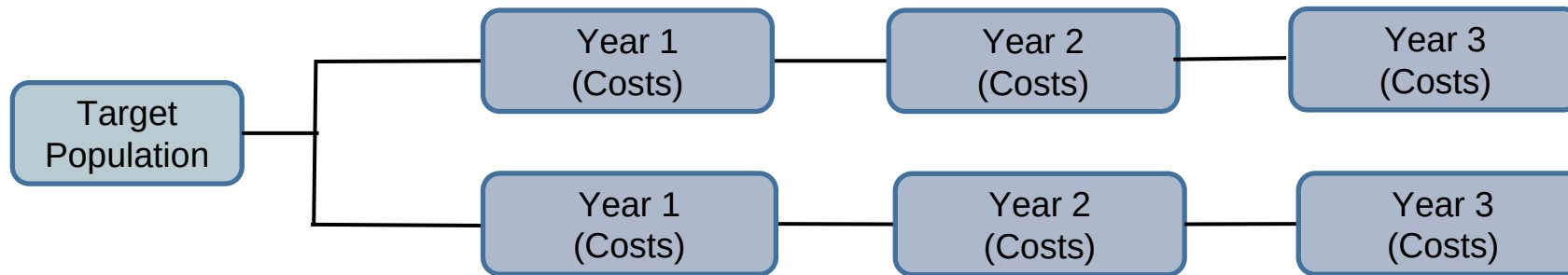
Costs examined?



Outcomes examined?



BIA considers costs a little differently to the other economic evaluation methods



Key  
Feature:

BIA does not report outcomes and is only concerned with the change in expenditure. BIA also accounts for target population size, patient eligibility, speed of uptake and replacement effects of the new intervention.



## SACMC examples: Budget impact analyses



### 1. National COVID-19 Budget Model

Cost of COVID-19 care and required quantities (ICU/ general beds, staff, oxygen, ventilators, drugs, PCR tests and testing algorithms, field hospitals, mortuary containers)

→ *COVID-19 Adjustment Budgets (nat'l and prov'l)*



### 2. COVID-19 Vaccine Cost Model

Types and quantities of vaccines (incl. boosters), size of target populations

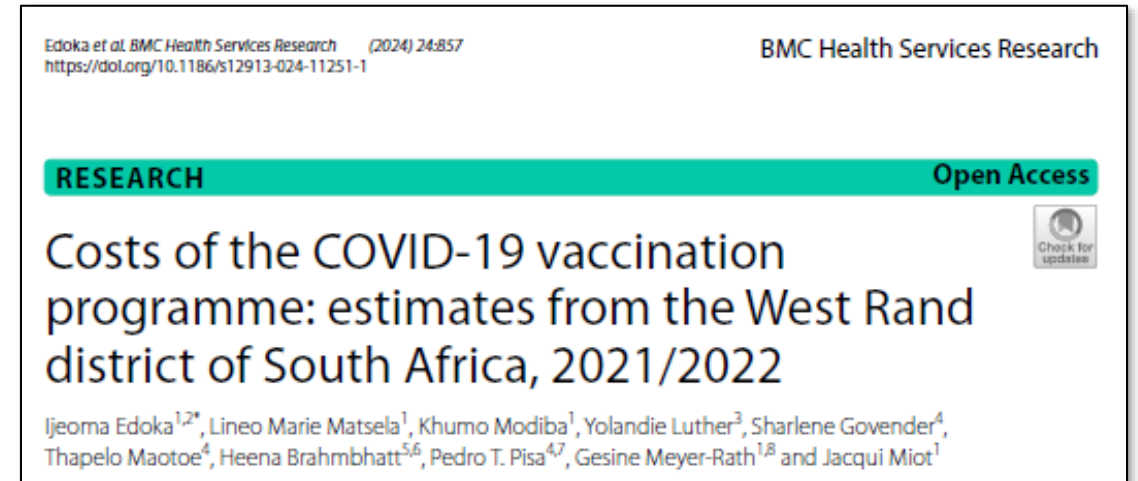
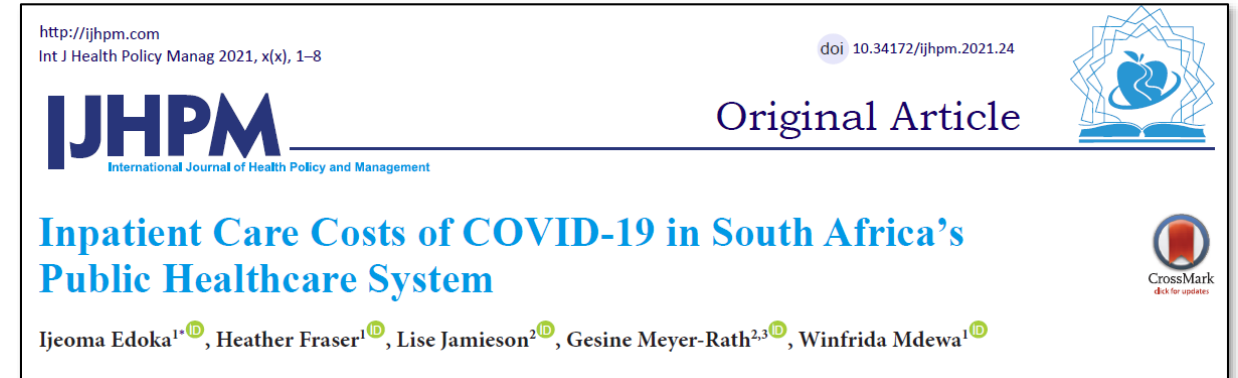
→ *Vaccine budgets (nat'l and prov'l), size of vaccine orders*



## SACMC examples: Cost analyses

### Cost analyses of

- hospitalisation costs (prospective + retrospective)
- vaccination programme (prospective + retrospective)

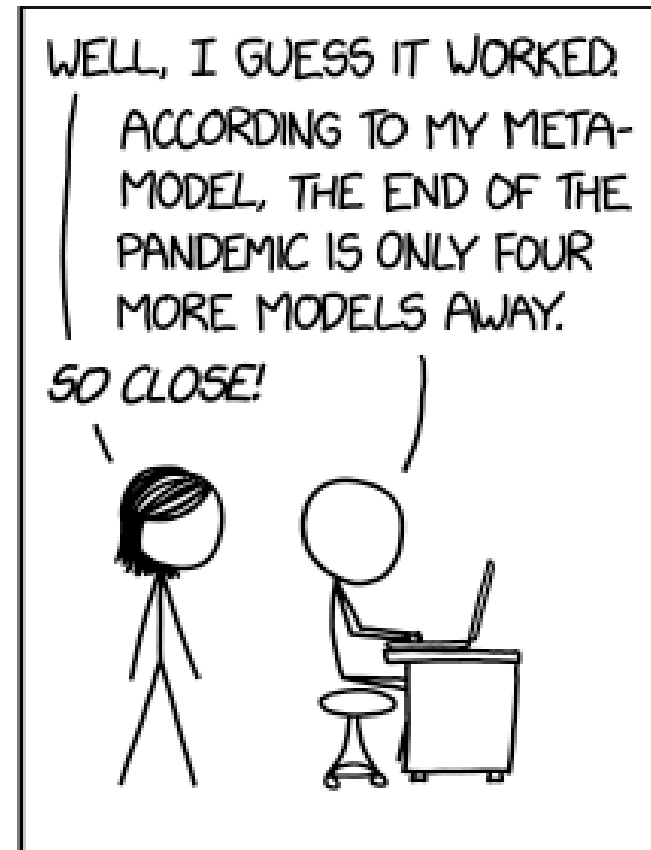
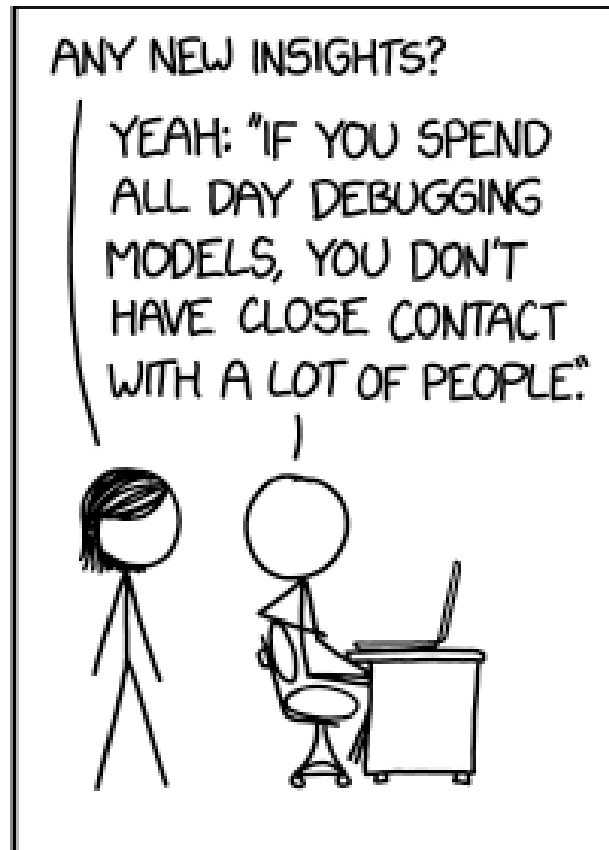


# Practical I

1. Working with a partner, choose one (or more) of the following **decision problems**:
  1. Should Ghana pay to roll out the newly approved MVA-BN mpox vaccine in the public sector?
  2. Is it more important to spend our health budget on infectious diseases or on NCDs?
  3. A new diabetes drug has the same effectiveness as the old one. Should we buy it?
  4. Should the government budget be spent on hiring more ECD educators, primary school educators or nurses?
2. Together, jot down the answers to the following questions:
  - What **type of analysis** will you do to answer the question?
  - What **data** will you need- a) costs and/ or b) outcomes?
  - If you need outcome data, what **type of outcome**?

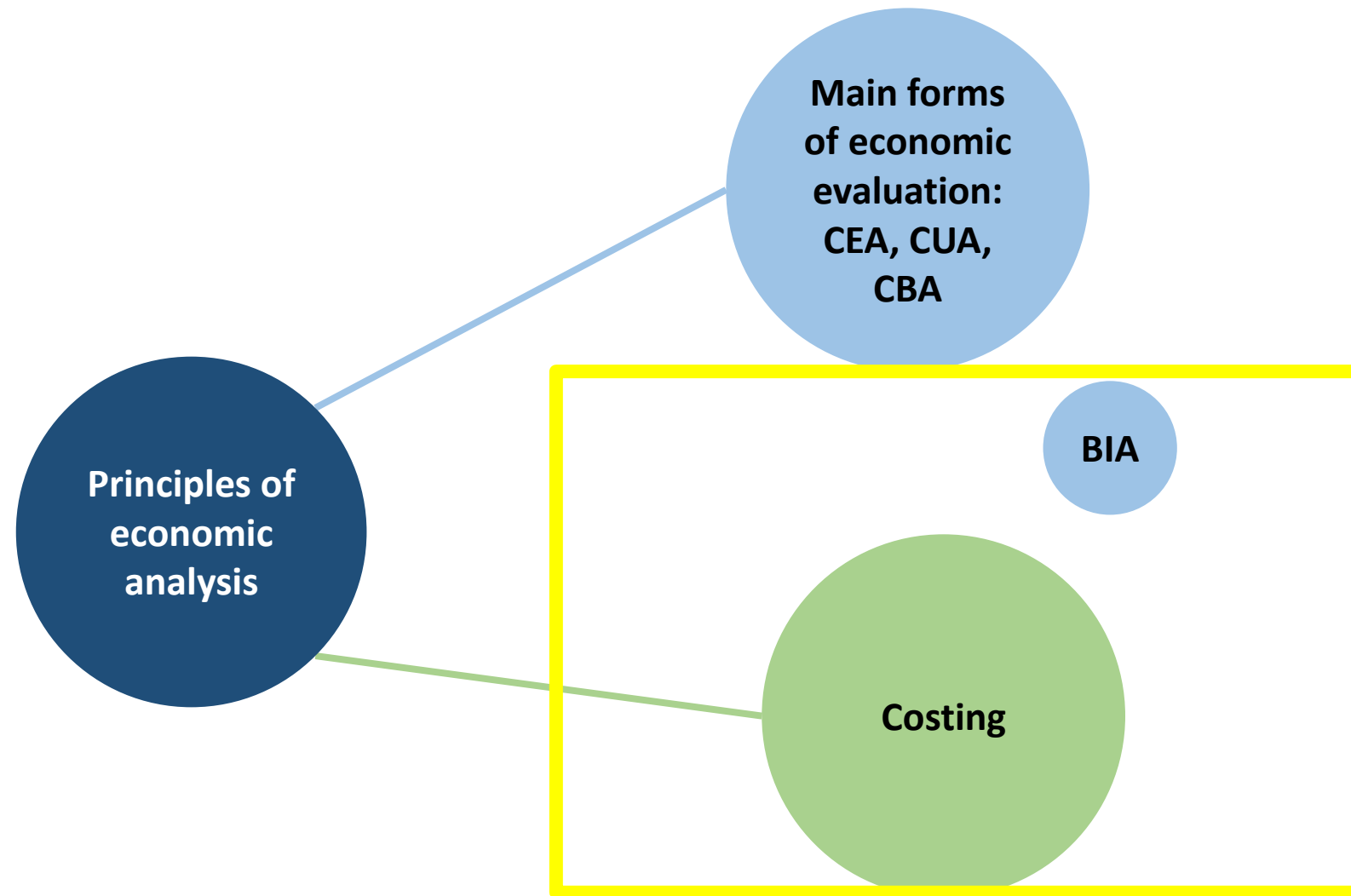
# Practical I - Answers

|                                                                                                         |                  | Data needed |           |                 |
|---------------------------------------------------------------------------------------------------------|------------------|-------------|-----------|-----------------|
| Decision problem                                                                                        | Type of analysis | Cost?       | Outcomes? | Which outcomes? |
| Should Ghana pay to roll out the newly approved MVA-BN mpox vaccine in the public sector?               |                  |             |           |                 |
| Is it more important to spend our health budget on infectious diseases or on NCDs?                      |                  |             |           |                 |
| A new diabetes drug has the same effectiveness as the old one. Should we buy it?                        |                  |             |           |                 |
| Should the government budget be spent on hiring more ECD educators, primary school educators or nurses? |                  |             |           |                 |





# Focus of this session



# What is a cost?

The economic concept of **cost** is based on opportunity cost

## Opportunity cost:

the *foregone value/ benefit* of the next best alternative

- What is the value of your course attendance this week?



# Types of cost: Capital vs recurrent costs

## Capital costs:

- Items with long lives (often defined as **over one year**)
- Incurred with large time intervals (e.g., new vehicles)
- Sometimes use cost minimum (e.g. \$100)
- Calculated by dividing cost of item by its useful life span (eg, 5 years for laptop, 10 years for vehicle)



## Recurrent costs:

- resources that produce benefits for one year or less
- Incurred on a continual basis (e.g., rent, salaries)



Key  
Feature:

### Main difference:

The length of time a resource is expected to produce benefits, or the interval at which costs will be incurred

# Types of cost: Cost measures

- **Total cost (TC):**

- cost of producing a quantity of services or outputs for a project
- presents the cost of all resources used in the intervention
- sum of all costs incurred (resources used) over the specified period of the analysis

- **Average cost (AC)** (or unit cost):

- Total Cost per unit of output
- cost per visit, cost per person vaccinated, *etc.....*

$$\text{Average cost} = \frac{\text{Total cost}}{\text{Units of output}}$$

# Identifying, measuring and valuing costs

1. **Identify** all steps included in the intervention (and how it differs from standard practice/ each comparator)  
→ **ingredients**
2. Decide on the most relevant sources to **measure** relevant resource use  
→ **quantities, Q**
3. Decide on the best method to elicit the **value** of each resource, relevant to the perspective chosen  
→ **prices, P**

## Ingredient

Nurses

Volunteers

Bikes

Cool boxes

Vials

# IDENTIFY: Understanding the intervention

- First step in cost analysis is to understand how an intervention works and what is being done:
  - community- or facility-based
  - public or private
  - activities: demand creation, outreach, M+E , supply chain
- Ask/ define:

**What?**  
**Where?**  
**Why?**  
**By whom?**

**When?**  
**How?**  
**How often?**  
**For whom?**

**Where does my intervention start?**  
**Where does it end?**

**...and who needs the results?**

# IDENTIFY: Vaccination for new pathogen

- First step in cost analysis is to understand how an intervention works and what is being done:
  - community- or facility-based **BOTH**
  - public or private **BOTH**
  - activities: demand creation, outreach, M+E, supply chain **ALL OF THESE**
- Ask/ define:

**What?**  
**Where?**  
**Why?**  
**By whom?**

**When?**  
**How?**  
**How often?**  
**For whom?**

**Where does my intervention start?**  
**Where does it end?**

**...and who needs the results?**

# **MEASURE:** Decide on quantities (resource use)

- **Collect data directly from patients or providers**
  - Data collection from sample of patient files
  - If collecting from trial, make sure you adjust for trial-induced resource use in a 'routine care' scenario
    - No safety labs, less visits, potentially lower staff cadres
    - Consider adjusting outcomes to represent less stringent care level
- **Identify relevant work done** previously which can be adapted to your analysis (**literature review**)
  - Adapt for relevant (local/ current) model of care/ visit frequency/ staff types etc
- **Use administrative data**
  - Public sector: DHIS: number of vaccinations; contraceptive coverage, etc
  - Private sector: Medical aid claims, etc
- **Use expert input**



# VALUE: Attach price levels

- **Collect data directly from same setting as resource use and outcomes**
  - Expenditure records
- **Identify relevant previous work** which can be adapted to your analysis **(literature review)**
  - Adapt for relevant (local/ current) input prices, salaries etc
  - If no access to current input prices, adjust for inflation
  - If no access to local input prices, convert from other currencies
- **Use administrative data**
  - Public sector:
    - Government tender records
    - DHIS: per patient day equivalent
  - Private sector: Medical aid claims, etc
- **Use expert input**

# Organising cost results

- **By cost category:**
  - building, equipment, vehicles, personnel, supplies, overheads, etc.
- **By function or activity:**
  - demand creation, vaccination, M + E, etc
- **By organisational level:**
  - national, district, community
- **By source of funding:**
  - government, donors, NGOs, patient-fees, etc.

# Allocation of joint resources

- Often resources are shared by a number of programmes
  - e.g., office space, vehicle, nurses' time
- Need to estimate how much of their value is used in the specific programme
- Apportion by using the main cost driver (as % of total)

| INPUT     | COST DRIVER (= allocation factor)         |
|-----------|-------------------------------------------|
| Vehicles  | Distance travelled / time used            |
| Equipment | Time used / output produced (e.g. X-rays) |
| Building  | Time used / space used                    |
| Personnel | Time worked                               |
| Supplies  | Quantity used                             |

# Cost categories: Vaccination for new pathogen

| Category | Type of cost | Unit | Quantity | Source of price data |
|----------|--------------|------|----------|----------------------|
|----------|--------------|------|----------|----------------------|

Capital



Recurrent



# Practical II

Working with a partner, copy the table, choose a country, and fill out the remaining columns

- For capital items such as supply chain and M+E system, add “System” as unit, and “1” as quantity

# Cost categories: Vaccination for new pathogen

| Category                                                                                                                                                                                                                                                                                                                                                                         | Type of cost                | Unit                     | Quantity | Source of price data |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------------------|----------|----------------------|
| <b>Capital</b><br><br><br>                                                                                                                                                                                     | <b>Supply chain</b>         | System                   | 1        | MoH                  |
|                                                                                                                                                                                                                                                                                                                                                                                  | <b>M+E system</b>           | System                   | 1        | MoH                  |
|                                                                                                                                                                                                                                                                                                                                                                                  | <b>Training</b>             | Per vaccinator           | 1        | MoH                  |
|                                                                                                                                                                                                                                                                                                                                                                                  | <b>Demand creation</b>      | Per vaccinee             | 1        | MoH                  |
|                                                                                                                                                                                                                                                                                                                                                                                  | <b>Equipment: - Fridges</b> | Per clinic               | 1        | Tender               |
|                                                                                                                                                                                                                                                                                                                                                                                  | - Cooler boxes              | Per outreach site        | 5        | Tender               |
|                                                                                                                                                                                                                                                                                                                                                                                  | - Sharps containers         | Per vaccinator           | 1        | Tender               |
|                                                                                                                                                                                                                                                                                                                                                                                  | <b>Vaccination cards</b>    |                          |          | MoH                  |
| <b>Recurrent</b><br><br><br><br><br> | <b>Staff</b>                | Per vaccination (mins)   | 15       | Salary tariffs       |
|                                                                                                                                                                                                                                                                                                                                                                                  | <b>Vaccine vials</b>        | Per vaccinee             | 1/4      | Vaccine contracts    |
|                                                                                                                                                                                                                                                                                                                                                                                  | <b>Wastage</b>              | Per vaccinee             | 10%      | Literature           |
|                                                                                                                                                                                                                                                                                                                                                                                  | <b>Syringes</b>             | Per vaccinee             | 1        | Tender               |
|                                                                                                                                                                                                                                                                                                                                                                                  | <b>Alcohol swabs</b>        | Per vaccinee             | 1        | Tender               |
|                                                                                                                                                                                                                                                                                                                                                                                  | <b>Overhead</b>             | Per site/ m <sup>2</sup> | 1        | Literature           |

# Costing for budget impact analysis

- **Economic analysis** answers the question “**Should we do it?**”
    - often compared to other options
    - always includes outcomes (LY, QALYs, DALYs, etc)
  - **Budget impact analysis** answers the question “**Can we afford it?**”
    - compared to an existing budget
    - does not include outcomes (but includes cost impact)
- Affordability should never be the first, but will often be the *second* question to ask before implementing a new intervention
- Budget impact analysis comes AFTER a decision for a new intervention has been made based on economic evaluation

# Budget impact analysis: Two approaches

- **Simple approach**

- Estimate the number of patients using product (**A**)
- Estimate the unit cost of product (**B**)
- Estimate the rate of increase in uptake over time (e.g. 3-5 years) e.g. 5% increase per year, or scale up to 90% coverage over time

$$\text{BUDGET} = \text{UNIT COST} * \text{TARGET POPULATION} * \text{COVERAGE}$$

- **Comprehensive approach/ Dynamic model**

- Allow eligible population size and disease severity to change over time



# Budget impact analysis: Principles

- Know the budget holder (government, donor)
- Budget holder defines **perspective**, which **cost items** to include and **time frame** (= budget period)
- Include all **follow-on and replacement effects** from introducing new technology
- **Capital costs are not annualised** (so budget can be allocated to years when needed)
- Limited sensitivity analysis (**scenario analysis** only); validation with decision makers

# Budget impact analysis: Things to consider

## Substitution/ Replacement

- Whether intervention is replacing/ decreasing existing standard of care (*eg, HIV self tests replacing clinic-based HIV rapid tests*)

## Combination

- Whether the intervention is being used in addition to the existing standard of care (*eg, injectable PrEP in addition to injectable contraception*)

## Expansion

- Whether the intervention will be used in a situation where there has been no existing care (*eg, COVID-19 tests*)

## Follow-on

- Whether intervention will increase use of another service (*eg, improved diabetes screening increasing diabetes treatment need*)

# Further readings

- **Costing principles:**
  - Global Health Cost Consortium (GHCC) Reference Case ([https://ghcosting.org/pages/standards/reference\\_case](https://ghcosting.org/pages/standards/reference_case))
- **BIA guidelines:**
  - Sullivan SD, et al: Value Health. 2014 Jan-Feb;17(1):5-14. doi: 10.1016/j.jval.2013.08.2291.
- **Generic cost sources:**
  - WHO CHOiCE database
  - GHCC unit cost database
  - **YOUR OWN MOH**
- **UNICEF/ WHO COVID-19 vaccine cost tool:** [https://www.who.int/publications/i/item/who-2019-ncov-vaccine\\_deployment\\_tool-2021.1](https://www.who.int/publications/i/item/who-2019-ncov-vaccine_deployment_tool-2021.1)

